

Car Rental Management System

Research Methodology

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Abstract-

The research was conducted to carry out the manual management system into a computer-based system for car rental services. so, the methodology chosen for this research paper was explanatory and correlational. this section describes the variables used in this research, the approach used to perform the research, research layout, characteristics of the tools used for data collection, data collection practices, target population, statistical techniques, etc..

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Introduction

The existence system has many problems in the working procedure of the company. Car rental management system is very effective, efficient, reliable and economic from all aspects. It provides all the current needs of the user when it comes to car rental services. It is managed details efficiently and smoothly. The purpose of this system is to automate the existing manual system by the help of computerized equipment and full-fledged computer software, fulfilling their requirements, because of that their valuable data/ information can be stored for a longer period with easy accessing and manipulation of the same. The required software and hardware are easily available and easy to work with.

Evaluation of Design Techniques

Evaluating alternative solutions involves analysis, discussion, and review. Iterative cycles of analysis are sometimes necessary. Supporting analyses, experimentation, prototyping, piloting, or simulations may be needed to substantiate scoring and conclusions.

Table 1- Design Techniques

Project cost	Available resources	Legal Requirements
: Basically, two main divisions: - Development costs of developing and implementing the new solution and, for an IT solution. Ex: cost for computer devices. - Operating costs of running the new system. Ex. Cost for training staff members.	This includes the hardware and software resources that are available for product development .	Have obtain license key for servers used

Decisions are taken during the Design process

-Collect of data has been done by conducting interviews and questionnaires from the members of car rent service.

-To complete this project, use the Visual studio 2019 software and C# programming language technology basically.

System Design Process

In this section provides a mobile phone repair management system's diagram-based design. Under this section Use Case Diagrams, high-level system architectures will describe.

Use Case Diagrams

A use case diagram at its simplest is a representation of a user's interaction with the system that shows the relationship between the user and the different use cases in which the user is involved. A use case diagram can identify the different types of users of a system and the different use cases and will often be accompanied by other types of diagrams as well. The major purpose of the use case diagram is the identify boundary of the system and it is a representation of a user's contribution to the system. The following use cases have been identified for the proposed system specification.

- Login Management
- Car Management
- Customer Management
- Booking Management
- Payment Management
- Staff Management
- Maintenance of vehicle

Actor: An actor represents human users interacting with the system and the actor can participate in one or more use cases.

Table 02- use case for login

Use Case Name	Login Management
Description	Users and admins can login to the system
Actor	Users/admin

*Table 2***Table 03- use case for Car management**

Use Case Name	Car management
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Description	Adding new cars and details to the system
Actor	Users/admin
Main -Success Scenario	System should allow to user can register a new car. System should allow to save cars' details in database. System should allow to view, edit and delete cars' details

Table 3

Table 04- use case for Customer Management

Use Case Name	Customer Management
Description	Adding new customers to the system
Actor	Users/admin
Main -Success Scenario	System should allow to user can register a new customer. System should allow to save customers' details in database.
Alternative Scenario	System should allow to view, edit and delete customers' details

Table 4

Table 05- use case for Booking management

Use Case Name	Booking management
Description	Adding booking details

Actor	Users/admin
Main -Success Scenario	System should allow to enter booking information's and save the database. System should allow to view, edit and delete booking details.

Table 5

Table 06- use case for Staff management

Use Case Name	Staff management
Description	Maintain Staff members details
Actor	admin
Main -Success Scenario	User can add, view, update, delete staff information and save the database

Table 6

Table 07- use case for Maintenance of the vehicles.

Use Case Name	Maintenance of the vehicles
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Description	Maintain service details of cars
Actor	Users/Admin

Table 7

System Architecture

The following 1.0 figure shows the overall system’s high-level system architecture.

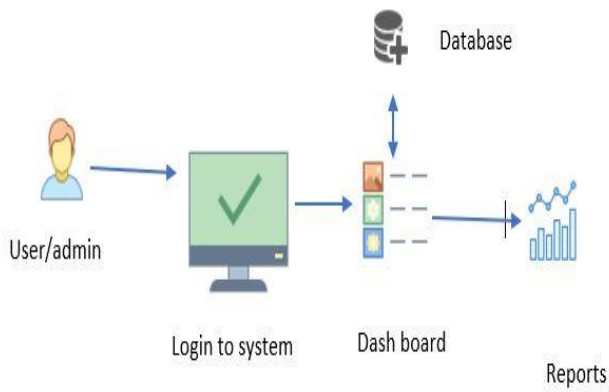
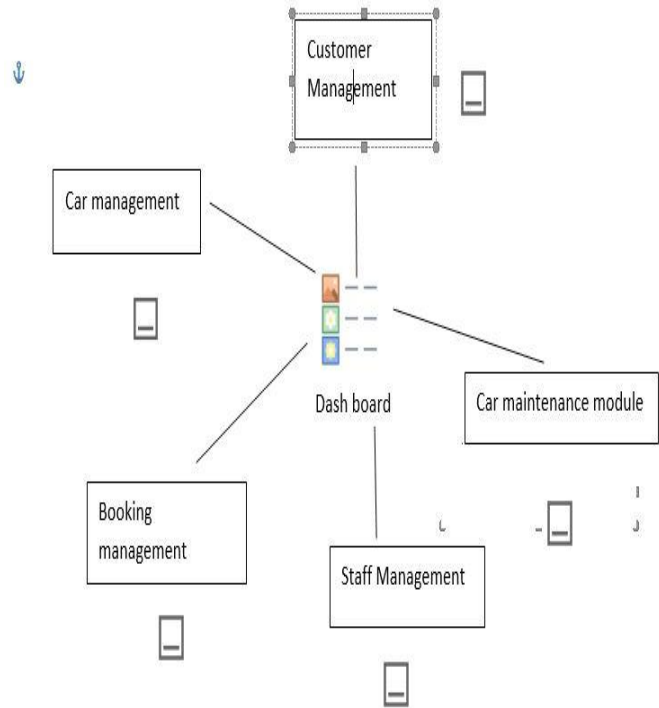


Figure 1

The following figure 2 shows the high-level architecture of the individual component.

Figure 2

Interface Design



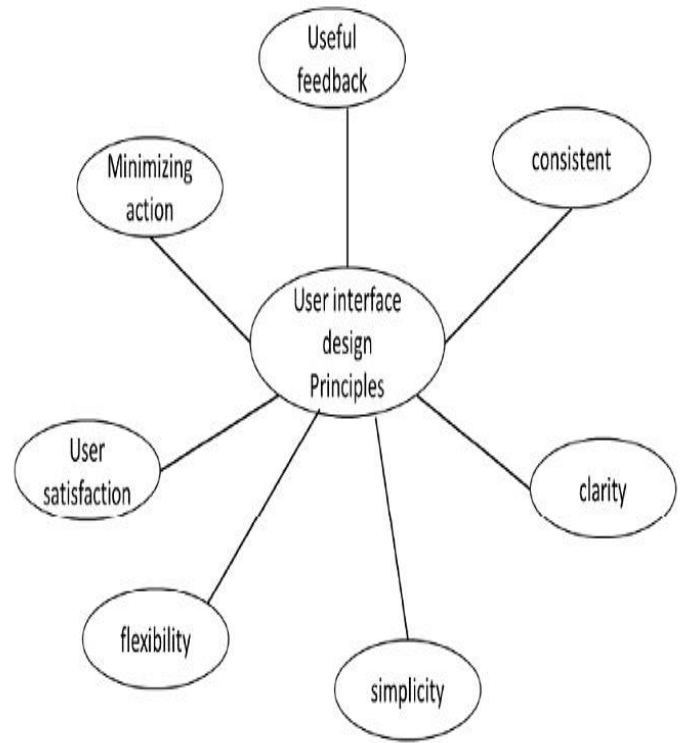
Any technique or method through which a product's end-user interacts with, or controls, a product, software, or hardware device is referred to as a user interface (UI). A good user interface should make the interaction between the user and the product as seamless, effortless, and enjoyable as possible. The user interface is meant to allow humans to control machines effectively and efficiently.

Good user interface design makes it easier to complete the work at hand without attracting too much attention to itself. Graphic design and typography are used to promote the interface's usability, affecting how the user performs particular interactions and boosting the design's aesthetic appeal, design aesthetics may increase or detract from users' ability to use the interface's functionalities. To produce a system that is not only operational but also accessible and adaptive to changing user needs, the design process must balance technical functionality with visual components.

Principles of User Interface Design

Now let us discuss the principles of user-interface design which are as follows –

- **Minimize actions** – Minimize action means steps per screen. The tasks and actions are streamlined so that they can be done in as few steps as possible. The interface should be designed keeping in mind to maintain the steps as few as possible for performing any tasks.
- **Simplicity** – The user interface should not be complex. It should always be designed simple and elegant.
- **Consistent** – The user interface should be consistent. The design should be consistent. Increasing consistency increases the familiarity and hence increases the usability.
- **Providing useful feedback** – The user should be provided with feedback for every action. This keeps the user informed and helps them to know whether some action was successful or not.
- **Clarity** – Content should provide the user with clarity. There should not be anything that confuses the user, as it becomes an obstacle for the user is interacting with the product.



Real Screen of the system

Login

Form1

User Name

User Name

Password

Clear

Login

Exit

Menu

Customers

Dashboard

Car management

Show Cars

Customer

Booking Management

Payment

Staff Management

Maintenance Of Vehicles

Income Viewer

Change Password

Customer Name

Contact Number

Customer Address

Customer NIC

Add

Update

Delete

Clear

Search Details

Show All

List Number	Name	Address	Contact Number	NIC Number
*				

Dashboard

Menu

Dashboard

Dashboard

Car management

Show Cars

Customer

Booking Management

Payment

Staff Management

Maintenance Of Vehicles

Income Viewer

Change Password

Vehicles

Staff Members

Customers

List Number	Registration Plate	Brand	Model and Year	Transmission	Fuel Type	Price	Available
2	asd	sd	asd	asd	asd	40	No
7	+123	toyota	202	Auto	Deaf	150	No
3	oooooooo	toyota	2013	2013	2013	4000	Yes
6	ddd	dy	ddy	Manual	Deaf	101	No
9	HH	toyota	2016	Auto	Peel	150	Yes
1	sca	sd	asd	asd	asd	40	No

Customer management

Car Management

Menu

Car Management

Dashboard

Car management

Show Cars

Customer

Booking Management

Payment

Staff Management

Maintenance Of Vehicles

Income Viewer

Change Password

Registration Plate

Transmission

Price

Brand

Fuel Type

Available

Model and Year

Add

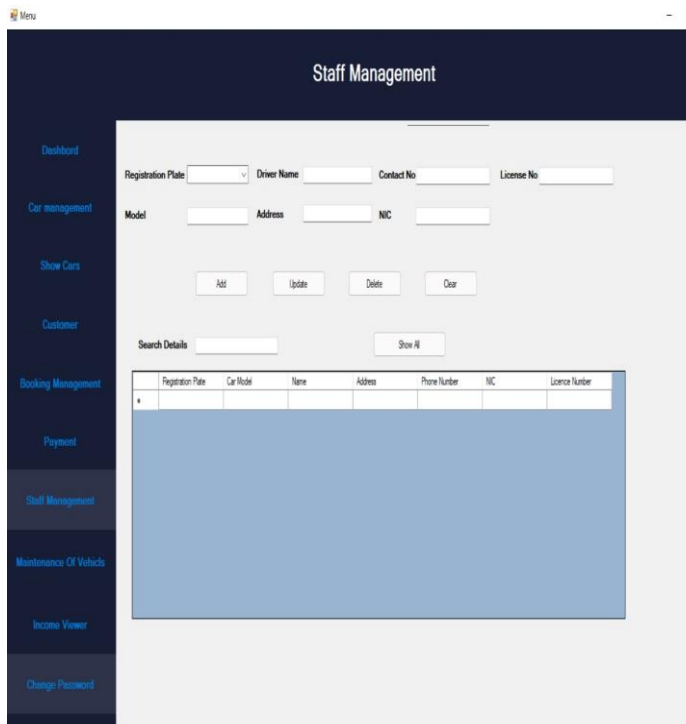
Update

Delete

Clear

List Number	Registration Plate	Brand	Model and Year	Transmission	Fuel Type	Price	Avail
*							

Staff Management



Database Design

Database design is the process of producing a detailed data model of a database. This data model contains all the needed logical and physical design choices and physical storage parameters needed to generate a design in a data definition language, which can then be used to create a database. A fully attributed data model contains detailed attributes for each entity.

Entity Relationship Diagrams (ERD)

An entity-relationship model also called an entity-relationship (ER) diagram, is a graphical representation of entities and their relationships to each other, typically used in computing in regard to the organization of data within databases or information systems. An entity is a piece of data-an object or concept about which data is stored.

Normalization

Normalization is the process of efficiently organizing data in a database. There are two goals of the normalization process: eliminating redundant data (for example, storing the same data in more than one table) and ensuring data dependencies make sense (only storing related data in a table). Both of these are worthy goals as they reduce the amount of space a database consumes and ensure that data is logically stored.

The Normal Forms

The database community has developed a series of guidelines for ensuring that databases are normalized. These are referred to as normal forms and are numbered from one (the lowest form of normalization, referred to as the first normal form or 1NF) through five (fifth normal form or 5NF). In practical applications, you'll often see 1NF, 2NF, and 3NF along with the occasional 4NF.

First Normal Form (1NF)

The first normal form (1NF) sets the very basic rules for an organized database:

- Eliminate duplicative columns from the same table.
- Create separate tables for each group of related data and identify each row with a unique column or set of columns (the primary key).

Second Normal Form (2NF)

The second normal form (2NF) further addresses the concept of removing duplicative data:

- Meet all the requirements of the first normal form.
- Remove subsets of data that apply to multiple rows of a table and place them in separate tables.
- Create relationships between these new tables and their predecessors through the use of foreign keys.

Third Normal Form (3NF)

The third normal form (3NF) goes one large step further:

- Meet all the requirements of the second normal form.
- Remove columns that are not dependent upon the primary key.
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DATA

The research is based on data from multiple servers, both quantitative and qualitative. In order to gather requirements from customers and the owner, an initial survey was conducted. To find the data, a questionnaire was supplied, which was then utilized to evaluate and obtain the data.

Moreover, All the gathered from both questionnaires and interviews were analyzed and used the decision-making. For the interview, we have selected a shop owner and a few employees from the shop. A questionnaire was presented to find the data, which was used to analyze and retrieve the data.